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PHY202 Exam 4 on chapter(s): 22

100% Excellent!

Name:__

These questions assess your understanding of the material from Chapter(s) 22 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. A proton moves at $5.7 \times 10^6\ \frac{m}{s}$ perpendicular to a magnetic field. The field causes the proton to travel in a circular path of radius 24. cm. What is the strength of the magnetic field?

ANSWER: 0.25 T ✓

$q = 1.6 \cdot 10^{-19}\,C$

$m = 1.67 \cdot 10^{-27}\,kg$

$v = 5.7 \cdot 10^{6}\,m/s$

$r = .24\,m$

$B = ?$

$$r = \frac{mv}{qB} \rightarrow B = \frac{mv}{qr}$$

$$B = \frac{(1.67 \cdot 10^{-27}\,kg)(5.7 \cdot 10^{6}\,m/s)}{(1.6 \cdot 10^{-19}\,c)(0.24\,m)} = 0.25\,T$$

2. A 0.15-kg baseball, pitched at 27. $\frac{m}{s}$ horizontally and perpendicularly to Earth's horizontal 48.-μT field, has a 13.-nC charge on it. What distance is the ball deflected from its path by the magnetic force after it travels 9.5 m horizontally? Do not consider the displacement due to the weight force. You may assume that the ball does not hit the ground during this motion.

ANSWER: $6.86 \cdot 10^{-12}\,m$ ✓

$m = 0.15\,kg$

$v = 27\,m/s$

$B = 48 \cdot 10^{-6}\,T$

$q = 13 \cdot 10^{-9}\,c$

$d = 9.5\,m$

$$F = qvb \sin\theta$$

$$F = (13 \cdot 10^{-9}\,c)(27\,m/s)(48 \cdot 10^{-6}\,T)(1)$$

$$F = 1.68 \cdot 10^{-11}\,N$$

$$F = ma \rightarrow a = \frac{F}{m} = \frac{(1.68 \cdot 10^{-11}\,N)}{(0.15\,kg)}$$

$$a = 1.12 \cdot 10^{-10}\,m/s^2$$

$$v = \frac{d}{t} \rightarrow t = \frac{d}{v} = \frac{9.5\,m}{27\,m/s} = 3.5 \cdot 10^{-1}\,s = 0.35\,s$$

$$x(t) = \tfrac{1}{2} a t^2 = (\tfrac{1}{2})(1.12 \cdot 10^{-10}\,m/s^2)(0.35\,s)^2 = \boxed{6.86 \cdot 10^{-12}\,m}$$

3. Find the current that must run through a long, straight wire so that it produces a 10.-μT magnetic field at a distance of 8.9 cm from the wire.

ANSWER: 4.45 A ✓

$I = ?$

$B = 10 \cdot 10^{-6}\,T$

$r = 8.9\,cm = 8.9 \cdot 10^{-2}\,m$

$$B = \frac{\mu_0 I}{2\pi r} \rightarrow I = \frac{B 2\pi r}{\mu_0}$$

$$I = \frac{(10 \cdot 10^{-6}\,T)(2)(\pi)(8.9 \cdot 10^{-2}\,m)}{(4\pi \cdot 10^{-7}\,Tm/A)}$$

$$\boxed{I = 4.45\,A}$$

+3

4. Calculate the radius of curvature of the path of an electron having a velocity of $1.5 \times 10^7 \frac{m}{s}$ perpendicular to a magnetic field of strength $B = 0.48$ T.

ANSWER: $1.78 \cdot 10^{-4}$ m

$r = ?$

$q = 1.6 \cdot 10^{-19}$ C

$m = 9.11 \cdot 10^{-31}$ kg

$v = 1.5 \cdot 10^7$ m/s

$B = 0.48$ T

$$r = \frac{mv}{qB}$$

$$r = \frac{(9.11 \cdot 10^{-31} \text{ kg})(1.5 \cdot 10^7 \text{ m/s})}{(1.6 \cdot 10^{-19} \text{ C})(0.48 \text{ T})}$$

$$\boxed{r = 1.78 \cdot 10^{-4} \text{ m}}$$

5. What is the magnitude of the maximum force exerted on an aluminum rod with a 5.1-μC charge that you pass between the poles of a 0.54-T permanent magnet at a speed of 17. $\frac{m}{s}$?

ANSWER: $4.68 \cdot 10^{-5}$ N

$F = ?$

$q = 5.1 \cdot 10^{-6}$ C

$B = 0.54$ T

$v = 17$ m/s

$$F = qvb\sin\theta$$

$$F = (5.1 \cdot 10^{-6} \text{ C})(17 \text{ m/s})(0.54 \text{ T})(\sin 90°)$$

$$F = 4.68 \cdot 10^{-5} \text{ N}$$

6. The hot and neutral wires supplying DC power to a light-rail commuter train carry 930 A and are separated by 52. cm. What is the magnitude of the force between 340 m of these wires?

ANSWER: $1.13 \cdot 10^2$ N

$I = 930$ A

$r = 0.52$ m

$F = ?$

$l = 340$ m

$$\frac{F}{l} = \frac{\mu_0 I^2}{2\pi r}$$

$$\frac{F}{l} = \frac{(4\cancel{\pi} \cdot 10^{-7} \text{ T}\cdot\text{m/A})(930 \text{ A})^2}{(2)(\cancel{\pi})(0.52 \text{ m})}$$

$$\frac{F}{l} = 3.33 \cdot 10^{-1} \text{ N/m}$$

+3

$$\boxed{F = 1.13 \cdot 10^2 \text{ N}}$$

7. A wire carries a 10. A current toward the south. If a large magnet were placed directly to the west of the wire and emits a 2.4 T magnetic field perpendicularly towards the wire, calculate the force exerted on the wire per length of wire.

ANSWER: 24 N/m ✓

$I = 10 A$

$B = 2.4 T$

$\frac{F}{l} = ?$

$$\frac{F}{l} = IB\sin\theta$$

$$\frac{F}{l} = (10A)(2.4T)(\sin 90^\circ)$$

$$\frac{F}{l} = 2.4\cdot10^{1} N/m = \boxed{24 N/m}$$

8. Find the current that must run through a current-carrying circular loop so that it produces a 220-µT magnetic field at its center, given that the radius of the circular loop is 17. cm.

ANSWER: 59.5 A ✓

$I = ?$

$B = 220\cdot10^{-6} T$

$r = 17 cm = 0.17 m$

$$B = \frac{\mu_0 I}{2R} \rightarrow I = \frac{B2R}{\mu_0}$$

$$I = \frac{(220\cdot10^{-6} T)(2)(0.17 m)}{(4\pi\cdot10^{-7} Tm/A)}$$

$$\boxed{I = 59.5 A}$$

9. Suppose that you rub a glass rod with silk, placing a 28.-nC positive charge on it. Calculate the initial force on the rod due to the Earth's magnetic field, if you throw the rod with a horizontal velocity of 2.1 $\frac{m}{s}$ due west in a place where Earth's field is due north. Recall that the magnitude of Earth's magnetic field is 35. µT.

ANSWER: $2.1\cdot10^{-12} N$ ✓

$q = 28\cdot10^{-9} C$

$F = ?$

$B = 35\cdot10^{-6} T$

$v = 2.1 m/s$

$$F = qvb\sin\theta$$

$$F = (28\cdot10^{-9}C)(2.1 m/s)(35\cdot10^{-6}T)(1)$$

$$\boxed{F = 2.1\cdot10^{-12} N}$$

+3

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10. A large water main is 1.4 m in diameter and the average water velocity is 0.50 $\frac{m}{s}$. Find the Hall voltage produced if the pipe runs perpendicular to Earth's $5.0 \times 10^{-5}$-T field.

ANSWER: $3.5 \cdot 10^{-5}$ v

$d = 1.4$ m

$v = 0.50$ m/s

$\mathcal{E} = ?$

$B = 5 \cdot 10^{-5}$ T

$\mathcal{E} = B l v$

$\mathcal{E} = (5 \cdot 10^{-5} T)(1.4 m)(0.5 m/s)$

$\boxed{\mathcal{E} = 3.5 \cdot 10^{-5} v}$

11. A proton travels with speed $1.6 \times 10^{6}$ $\frac{m}{s}$ at an angle of 42.° to a uniform magnetic field. If its path has a circular component of radius 0.12 m, calculate the magnitude of the magnetic field.

ANSWER: $9.3 \cdot 10^{-2}$ T

$q = 1.6 \cdot 10^{-19}$ c

$m = 1.67 \cdot 10^{-27}$ Kg

$v = 1.6 \cdot 10^{6}$ m/s

$\theta = 42°$

$r = 0.12$ m

$B = ?$

$r = \frac{mv}{qB} \rightarrow B = \frac{mv \sin\theta}{qr}$

$$B = \frac{(1.67 \cdot 10^{-27} Kg)(1.6 \cdot 10^{6} m/s)(\sin 42°)}{(1.6 \cdot 10^{-19} c)(0.12 m)}$$

$B = 9.3 \cdot 10^{-2}$ T

12. Find the maximum torque on a 91.-turn square loop of wire that is 9.5 cm on each side that carries 3.4 A of current in a 1.5-T field.

ANSWER: 4.19 Nm

$T_{max} = ?$

$N = 91$

$l = 9.5 \cdot 10^{-2}$ m

$I = 3.4$ A

$B = 1.5$ T

$A = (9.5 \cdot 10^{-2} m)^2 = 9.03 \cdot 10^{-3} m^2$

$\theta = 90°$

$T = N I A B \sin\theta$

$T = (91)(3.4 A)(9.03 \cdot 10^{-3} m^2)(1.5 T)(1)$

$\boxed{T = 4.19 \text{ Nm}}$

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1. $B = 0.248\,\text{T} \because r = \frac{mv}{qB}$, $m_p = 1.67 \times 10^{-27}\,\text{kg}$, $q_p = 1.60 \times 10^{-19}\,\text{C}$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_p13)

2. $\Delta y = 6.95 \times 10^{-12}\,\text{m} \because F = qvb\sin\theta,\ F_{net} = ma,\ \vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$
(Chapter 22, Section (11) Applications of Magnetism - cp1e_ch22_p82)

3. $I = 4.45\,\text{A} \because B = \frac{\mu_0 I}{2\pi r}$
(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ch22_ex6)

4. $r = 1.78 \times 10^{-4}\,\text{m} \because r = \frac{mv}{qB}$, $m_e = 9.11 \times 10^{-31}\,\text{kg}$, $q_e = 1.60 \times 10^{-19}\,\text{C}$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_ex2)

5. $F = 4.68 \times 10^{-5}\,\text{N} \because F = qvb\sin\theta \therefore F_{max} = qvb$
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_p7)

6. $F = 113.\,\text{N} \because \frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$
(Chapter 22, Section (10) Magnetic Force betewen Two Parallel Conductors - cp1e_ch22_p50)

7. $\frac{F}{\ell} = 24.0\,\frac{\text{N}}{\text{m}} \because F = I\ell B\sin\theta,\ \theta = 90°$
(Chapter 22, Section (7) Magnetic Force on a Current-Carrying Conductor - cp1e_ch22_ex4)

8. $I = 59.5\,\text{A} \because B = \frac{\mu_0 I}{2R}$
(Chapter 22, Section (9) Magnetic Fields produced by Currents: Ampere's Law - cp1e_ch22_ex7)

9. $F = 2.06 \times 10^{-12}\,\text{N} \because F = qvb\sin\theta,\ \theta = 90°$
(Chapter 22, Section (4) Magnetic Field Strength - cp1e_ch22_ex1)

10. $\varepsilon = 3.50 \times 10^{-5}\,\text{V} \because \varepsilon = B\ell v$
(Chapter 22, Section (6) The Hall Effect - cp1e_ch22_p22)

11. $B = 0.0931\,\text{T} \because r = \frac{mv}{qB}$, $m_p = 1.67 \times 10^{-27}\,\text{kg}$, $q_p = 1.60 \times 10^{-19}\,\text{C}$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - tkd1)

12. $\tau = 4.19\,\text{N}\cdot\text{m} \because \tau = NIAB\sin\theta \therefore \tau_{max} = NIAB$
(Chapter 22, Section (8) Torque on a Current Loop: Motors and Meters - cp1e_ch22_ex5)